

By: AKH
 Rev.--- By: ---
 Chk: ---

Date Jun 2026
 Date ---
 Date ---

MacTest

Macaulay Beam Analysis - Example Worksheet

106 - (4) Span - UDL

Beam Sect = W14x61 W14x90
 A = 17.90 26.50 in²
 I = 640.00 999.00 in⁴

E = 29,000,000 lb / in²

Length Units in
 Force Units lb

Beam Segments (First Segment Begins at 0)

	Segment No.	X End	A	I	E
		in	in ²	in ⁴	lb / in ²
DO NOT DELETE ROWS (GRAPHED)	1	1200.00	17.90	640.00	2.900E+07
	2				
	3				
	4				
	5				

Supports

	Supp. No.	X	Type	Imposed Movement	
		in	---	* dY (vert)	** dr (rot)
DO NOT DELETE ROWS (GRAPHED)	1	0.0	P	0.0	0.000
	2	300.0	P	0.0	0.000
	3	600.0	P	0.0	0.000
	4	900.0	P	0.0	0.000
	5	1200.0	P	0.0	0.000

* (settlement) ** (Not Working)

Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

	Load No.	x1	x2	w1 (at x1)	w2 (at x2)
		in	in	lb / in	lb / in
DO NOT DELETE ROWS (GRAPHED)	1	0.0	1200.0	-5.1	-5.1
	2	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0
	4	0.0	0.0	0.0	0.0
	5	0.0	0.0	0.0	0.0

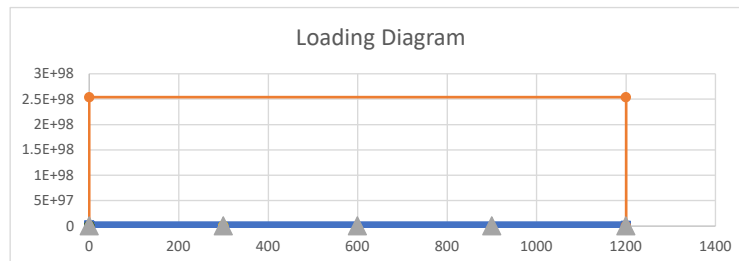
Load
lb / ft
61.0
0.0
0.0
0.0

Point Loads - UNFACTORED

	Load No.	X	Py	My
		in	lb	lb-in
DO NOT DELETE ROWS (GRAPHED)	1	300	0.00	0.00
	2	0	0.00	0.00
	3			
	4			
	5			

My Load
lb-ft
0.0
0.0
0.0

Loading Diagram



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BEAM REACTIONS

Position (X)	Enforced Disp (dy)	Enforced Rot (theta)	Vert Reac (Ry)	Internal Moment (Mz)	Type
(L)	(L)	(rad)	(F)	(F-L)	(---)
0.00	0.0	0.0	599.3	0.0	P
300.00	0.00	0.00	1,742.7	0.0	P
600.00	0.00	0.00	1,416.0	0.0	P
900.00	0.00	0.00	1,742.7	0.0	P
1,200.00	0.00	0.00	5.9931E+02	0.0000E+00	P

Vert Reac
(kip)
0.599
1.743

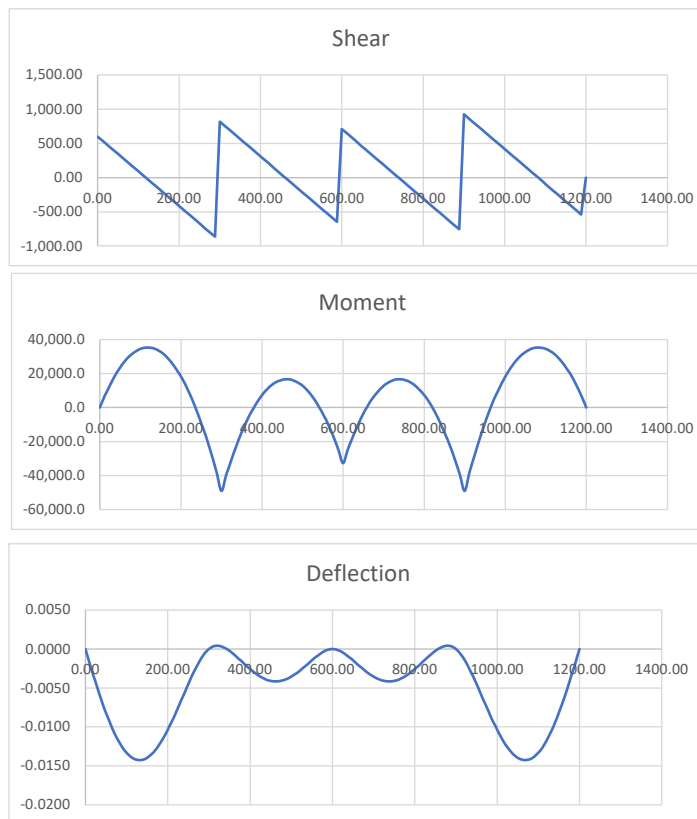
BEAM REACTIONS EQUILIBRIUM CHECK

Item	Units	Loads	Reactions	Residual	Check Sum
Sum Fy	F	-6,100.0	6,100.0	0.0	OK
Sum M @ x=0	F-L	-3.6600E+06	3.6600E+06	0.0	OK

BEAM MAX/MIN LOAD EFFECTS

Item	Position (X)	Shear (V)	Moment (M)	Defl. (delta)
(---)	(L)	(F)	(F-L)	(L)
Max +V	900.00	925.7	-48,956.2	0.0000
Max -V	288.00	-864.7	-38,214.0	-0.0007
Max +M	1080.00	10.7	35,317.5	-0.0141
Max -M	300.00	817.0	-48,956.2	0.0000
Max Defl	1068.00	71.7	34,823.3	-0.0143

Max/Min Value
(---)
0.93 kip
-0.86 kip
2.94 kip-ft
-4.08 kip-ft
-0.014 in



INPUT SUMMARY

General Info

Beam Length:	100 ft
Section Name:	W14x61
Self Weight:	False

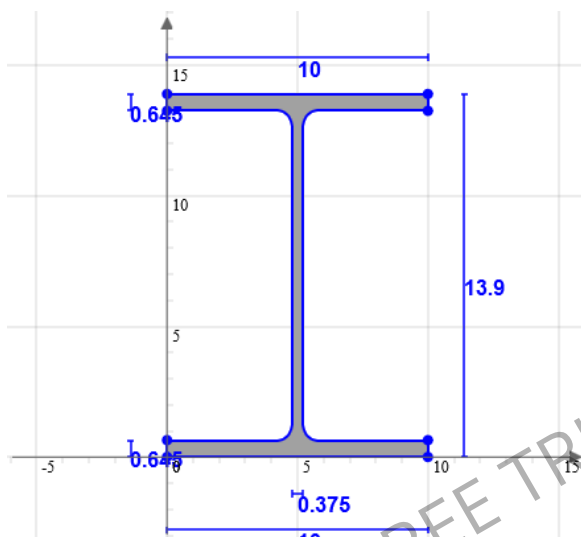
Supports

Support Type	Location
Pinned	0 ft
Roller	25 ft
Roller	50 ft
Roller	75 ft
Roller	100 ft

Loads

Load Type	Location	Magnitude	Load Case
Distributed Load	0 ft to 100 ft	-61 lb to -61 lb	DL

Beam Section



Geometric Properties		
A	17.9	in ²
C _z	5	in
C _y	6.95	in

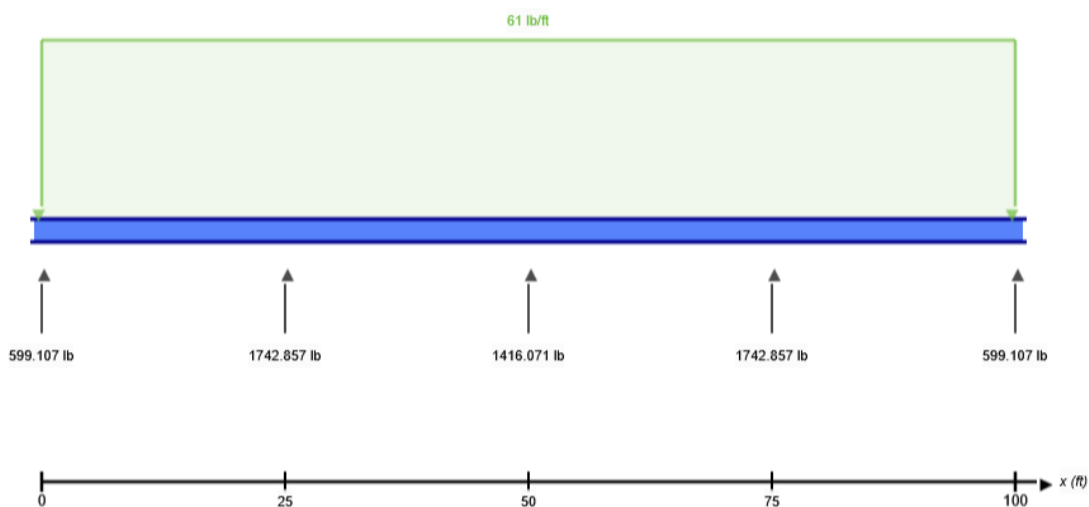
Bending Properties		
I _z	640	in ⁴
I _y	107	in ⁴

Shear Properties		
A _z	11.277	in ²
A _y	4.99	in ²

Torsion Properties		
J	2.19	in ⁴
r	0.991	in

Shape	Material	E (ksi)	v	ρ (lb/ft ³)
W14x61	Structural Steel	29000	0.27	490

FREE BODY DIAGRAM



RESULT SUMMARY

Check	Status	Limit	Ratio	Max
Deflection	PASS	L/250	0.012	L/21381
Custom Stress Limit	PASS	39 ksi	0.014	0.532 ksi
Material Yield	PASS	36 ksi	0.015	0.532 ksi
Material Strength	PASS	58 ksi	0.009	0.532 ksi

ANALYSIS RESULTS

Reactions

Support at	R_x (lb)	R_y (lb)	M (kip-ft)
0	0	599.107	0
25	0	1742.857	0
50	0	1416.071	0
75	0	1742.857	0
100	0	599.107	0

Force Extremes

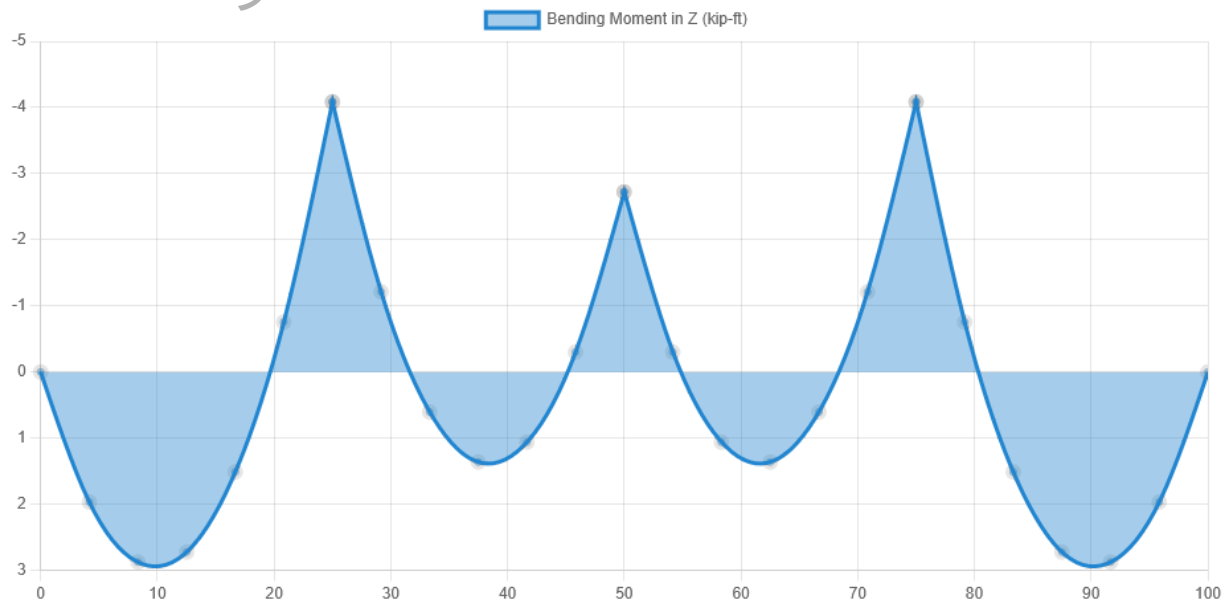
Result	Max	Min
Bending Moment	2.875 kip-ft	-4.085 kip-ft
Shear	925.893 lb	-925.893 lb
Displacement Y	0 in	-0.014 in

Stress Extremes

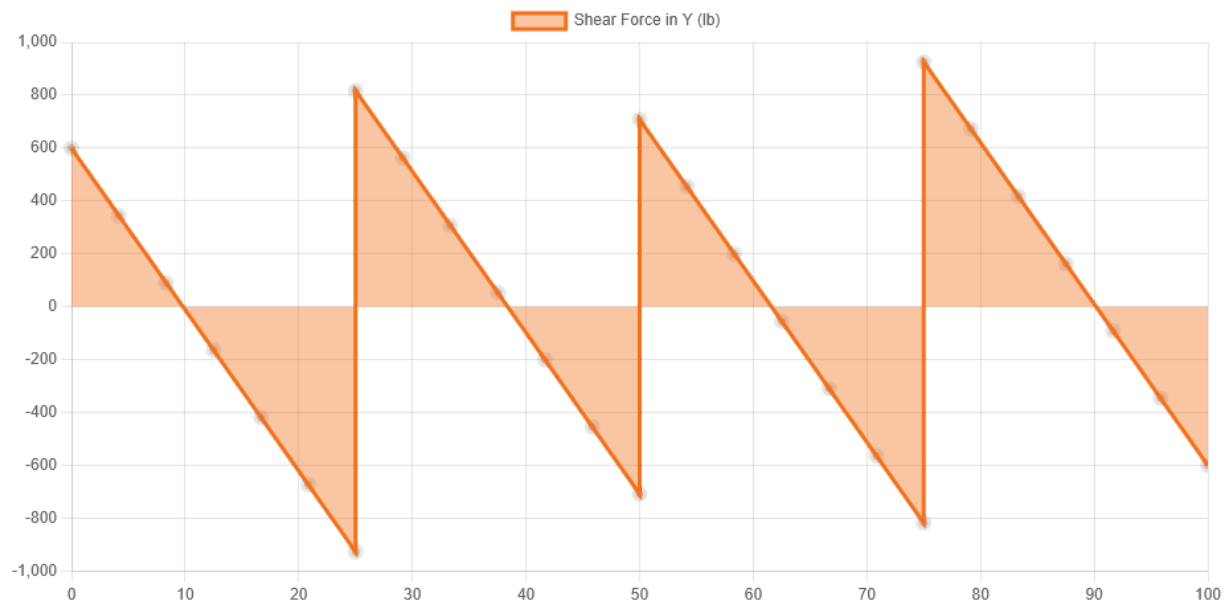
Result	Max (ksi)	Min (ksi)
Bending Stress	0.532	-0.532
Shear Stress Total	0.198	0.012
Max Combined Normal Stress	0.532	0
Min Combined Normal Stress	0	-0.532

DIAGRAMS

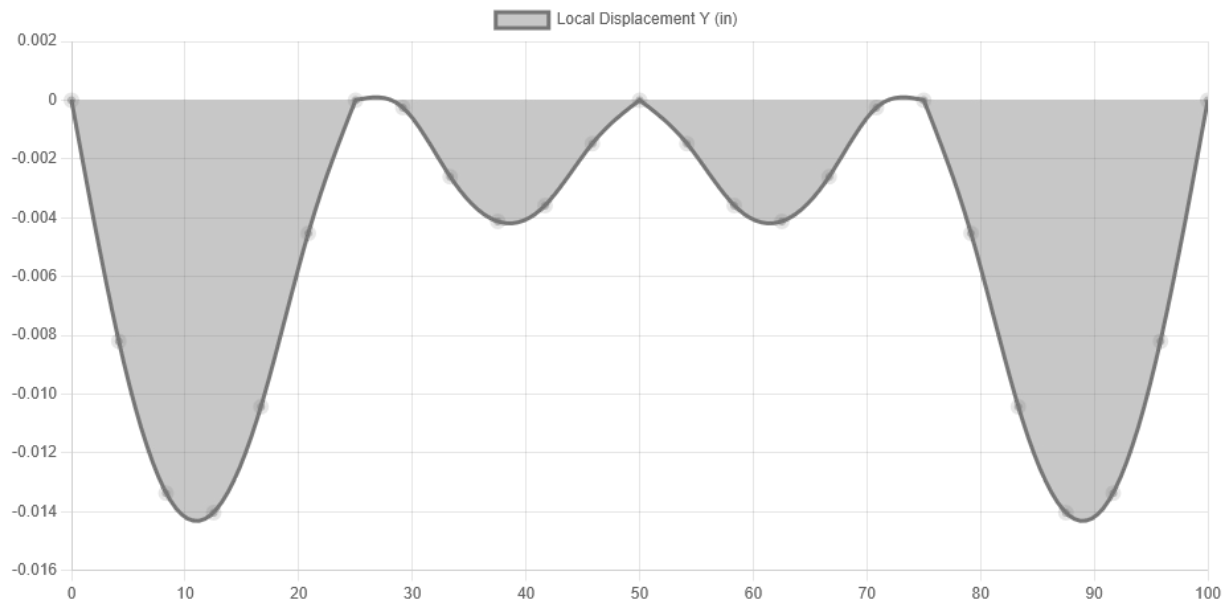
Bending Moment Diagram



Shear Force Diagram



Displacement



Location (ft)	Total Deflection (in)	Deflection/Span
0	0 ⓘ	-
12.5	0.014 ⓘ	L/21381
25	0 ⓘ	-
37.5	0.004 ⓘ	L/72698
50	0 ⓘ	-
62.5	0.004 ⓘ	L/72698
75	0 ⓘ	-
87.5	0.014 ⓘ	L/21381
100	0 ⓘ	-

ⓘ The Deflection/Span results are calculated using the analysis results and the Deflection Limit of L/250 set in the model settings.